

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (EE)

December 2013

EE308 High Voltage Engineering

Date: 05.12.2013, Thursday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 State and explain Paschen's law. Derive expressions for $(pd)_{min}$ and $(V_b)_{min}$. [07]
- Q - 2 (a) Enlist the different mechanisms observed in solid dielectrics and explain any two in detail. [10]
- (b) Define the following terms: [04]
- Dielectric Strength
 - Corona Discharge
 - Visual critical voltage
 - Disruptive Discharge Voltage

OR

- Q - 2 (a) Derive the criteria for the breakdown in electronegative gases. [08]
- (b) Explain the oil purification system in detail. [06]
- Q - 3 (a) Explain Cockcroft-Walton type voltage multiplier circuit in detail. [07]
- (b) An impulse generator has eight stages with each condenser rated for $0.16 \mu F$ and 125 kV. The load capacitor available is $1000 pF$. Find the wave front resistance and the wave tail resistance needed to produce $1.2/50 \mu s$ impulse wave. [04]
- (c) A generating voltmeter has to be designed so that it can have range from 20 to 200 kV dc. If indicating meter reads a minimum current of $2 \mu A$ and maximum current of $25 \mu A$, what should the capacitance of generating voltmeter be? Assume that the driving motor has a synchronous speed of 1500 rpm. [03]

OR

- Q - 3 (a) Enlist the methods used for the generation of high voltages and explain Marx impulse voltage generator in detail. [07]
- (b) A Cockcroft-Walton type voltage multiplier has eight stages with capacitances, all equal to $0.05 \mu F$, the supply transformer secondary voltage is 125 kV at a frequency of 150 Hz. If the load current to be supplied is 5 mA, find (a) the percentage ripple, (b) the regulation, and (c) the optimum number of stages for minimum regulation or voltage drop. [07]

SECTION – II

Q - 4 Define followings: [07]

- Dielectric loss • Puncture • Partial discharge
- Withstand voltage • Dissipation factor • Arc voltage
- Flashover

Q - 5 (a) Enlist the methods used for the measurement of high voltages and explain any one method in detail. [08]

(b) Explain the various factors affecting the high voltage measurement using sphere gaps. [06]

OR

Q - 5 (a) Explain the trigatron gap in detail. [06]

(b) Explain the various testing carried out on insulators. [08]

Q - 6 Explain any two from the followings: [14]

- (a) Capacitance voltage transformer
- (b) Testing of circuit breaker
- (c) High voltage Schering bridge
- (d) High voltage laboratory
